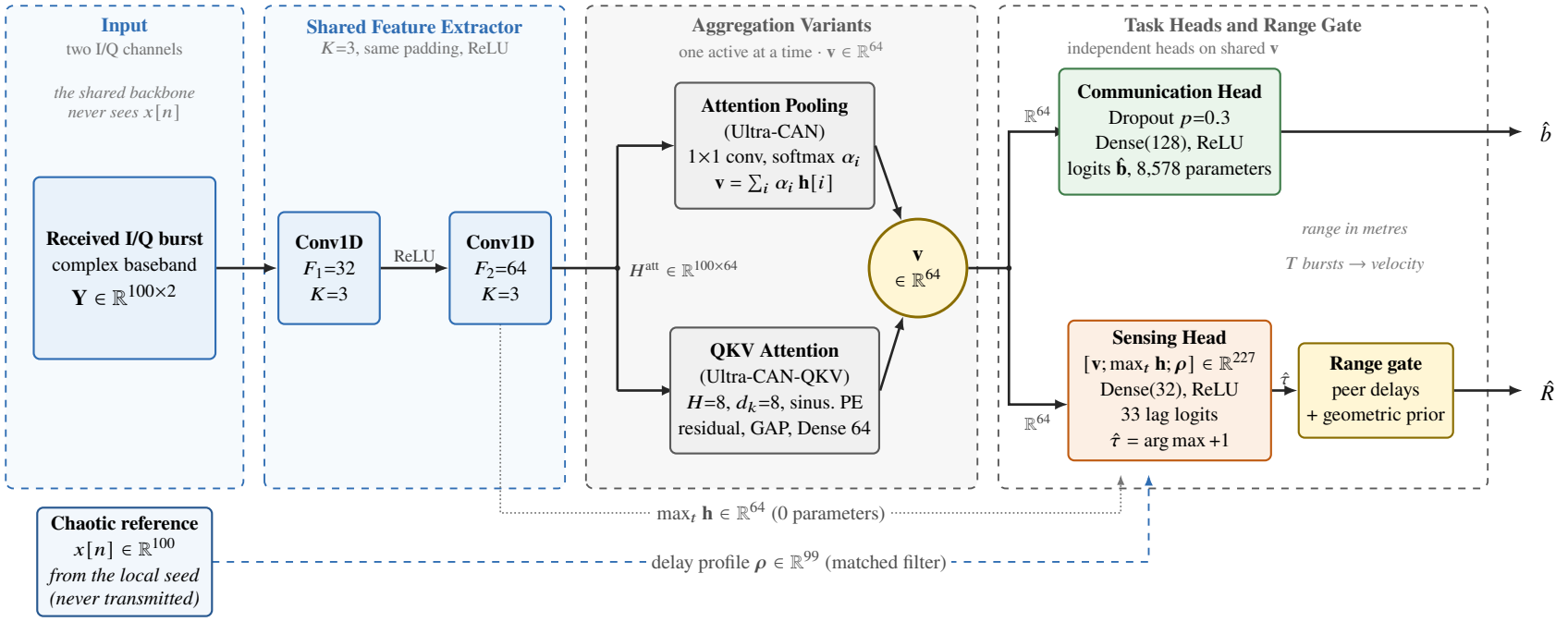

Input
two I/Q channels
the shared backbone never sees $x[n]$
Received I/Q burst
complex baseband
$\mathbf{Y} \in \mathbb{R}^{100\times2}$
Shared Feature Extractor
K=3, same padding, ReLU
Conv1D
F_1=32
K=3
ReLU
Conv1D
F_2=64
K=3
Aggregation Variants
one active at a time · $\mathbf{v} \in \mathbb{R}^{64}$
$H^{\text{att}} \in \mathbb{R}^{100\times64}$
Attention Pooling
(Ultra-CAN)
1×1 conv, softmax α_i
$\mathbf{v} = \sum_i \alpha_i \mathbf{h}[i]$
QKV Attention
(Ultra-CAN-QKV)
H=8, d_k=8, sinus. PE
residual, GAP, Dense 64
$\mathbf{v}$
$\in \mathbb{R}^{64}$
Task Heads and Range Gate
independent heads on shared $\mathbf{v}$
$\mathbb{R}^{64}$
Communication Head
Dropout p=0.3
Dense(128), ReLU
logits $\hat{\mathbf{b}}$, 8,578 parameters
$\hat{b}$
range in metres
T bursts → velocity
$\mathbb{R}^{64}$
Sensing Head
$[\mathbf{v}; \max_t \mathbf{h}; \boldsymbol{\rho}] \in \mathbb{R}^{227}$
Dense(32), ReLU
33 lag logits
$\hat{\tau} = \arg\max + 1$
$\hat{\tau}$
Range gate
peer delays
+ geometric prior
$\hat{R}$
$\max_t \mathbf{h} \in \mathbb{R}^{64}$ (0 parameters)
Chaotic reference
$x[n] \in \mathbb{R}^{100}$
from the local seed
(never transmitted)
delay profile $\boldsymbol{\rho} \in \mathbb{R}^{99}$ (matched filter)